

7. Simulation of Groundwater Flow

7.1 Basic equation of groundwater flow and flow net analysis

a) Darcy's law

According to Darcy's law, the velocity through porous media is proportional to hydraulic gradient.

$$v = -K(dh/dl)$$

where v = velocity of flow known as Darcy velocity or specific discharge; K = hydraulic conductivity; and dh/dl = hydraulic gradient

Discharge, $Q = Av = -KA(dh/dl)$

Darcy's equation in terms of transmissivity (T) is

$$v = -(T/H) \times (dh/dl) \quad [T = KH]$$

where H = thickness of aquifer

b) Steady continuity equation for incompressible flow

$$\partial u / \partial x + \partial v / \partial y + \partial w / \partial z = 0 \quad (I)$$

where u , v and w are velocity in x , y and z directions, respectively.

Stream line, equipotential line and flow net

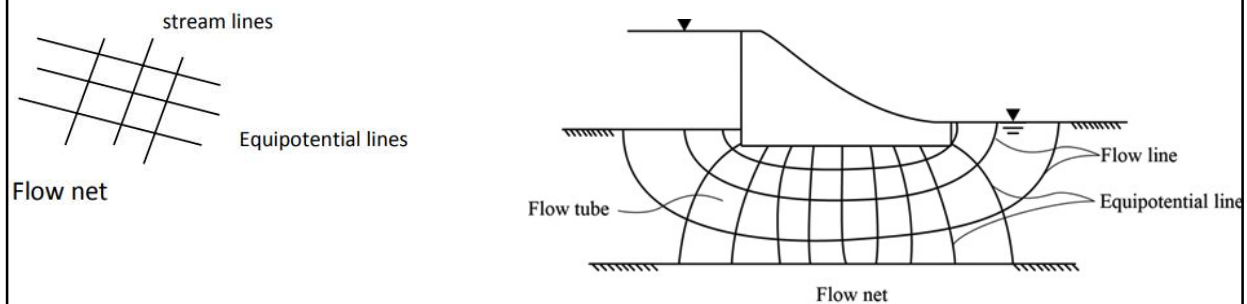
Stream line (flow line) is an imaginary curve drawn through the flow field in such a way that the tangent to it at any point indicates the direction of velocity vector at that point.

Along streamline, stream function is constant.

In steady flow, streamline coincides with the path line of flow.

An **equipotential line** is a line along which velocity potential (hydraulic head in groundwater flow) is constant.

A grid obtained by drawing a series of equipotential lines and stream lines is called a **flow net**.



Rules for constructing flow nets in isotropic medium

- Streamlines are drawn in the direction of flow; they can be curvilinear. They form a set of stream tubes. Streamlines cannot intersect.
- Equipotential lines are drawn perpendicular to the streamlines.

Total energy head or flow potential at any point on the water table is

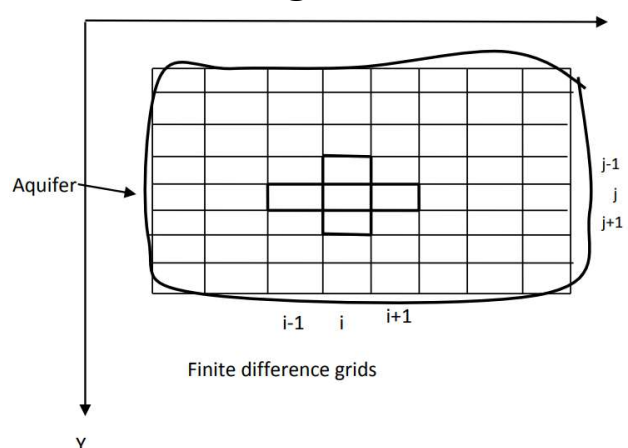
$$p/\gamma + z = h = \text{piezometric head}$$

Taking atmospheric pressure as datum, $h = Z$. Therefore, under steady conditions elevation at any point on the water table is equal to energy head and flow lines lie perpendicular to water table contours.

As contours of h represents ϕ , Darcy's equation can also be written as

$$v = -K(d\phi/dl) = -(T/H) \times (d\phi/dl)$$

7.2 Finite difference scheme for 2D groundwater simulation



Continuity equation for groundwater flow

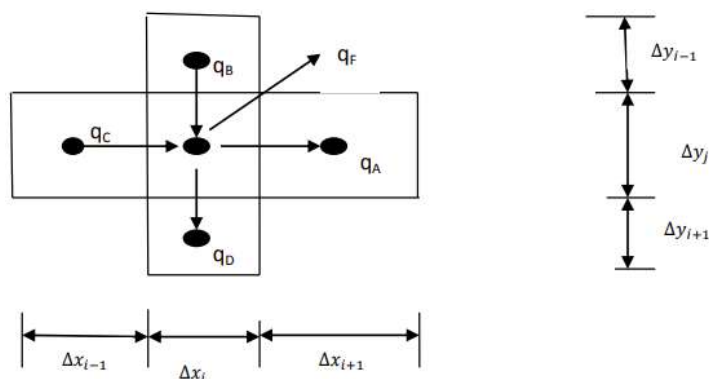


Figure: Enlarged view of interior grids defined by $i-1, i, i+1$ and $j-1, j$ and $j+1$

Flow takes place from center to center of grid.

q_A, q_B, q_C, q_D = grid to grid flow rates per unit area (Darcy flux)

q_F = net groundwater withdrawal rate per unit area from grid (i,j)

Continuity equation for grid (i, j)

inflow-outflow = rate of change of storage

$$\Delta x_i \Delta y_j [q_C - q_D - q_A + q_B - q_F] = S \frac{\partial \phi}{\partial t} \Delta x_i \Delta y_j \quad (I)$$

where S is the storage coefficient; ϕ is the potential head; and t is the time.

S = Volume of water released or taken / (area $\times$ change in GWT)

Rate of change of storage = $S \times$ rate of change of head $\times$ area

Darcy's equation in the form of ϕ (which is potential head h)

$$Q = -KA \frac{\partial \phi}{\partial x}$$

In terms of transmissivity

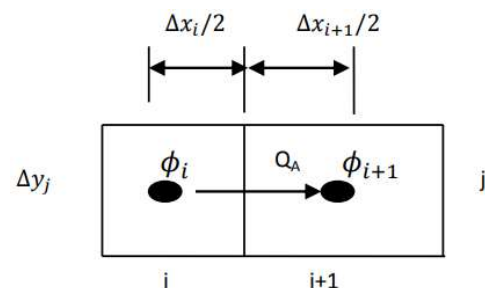
$$Q = -K \Delta x_i \Delta y_j \frac{\partial \phi}{\partial x} \quad (II)$$

Consider non-homogeneous and anisotropic aquifer.

In x - direction, $T_x = K_x \Delta x_i$

where T_x = weighted average transmissivity for grid i and $i+1$.

Consider (i,j) and $(i+1,j)$ grid



Finite difference form of Darcy's equation (II)

$$Q_A = -T_x(\Delta y_j) \frac{\phi_{i+1,j} - \phi_{i,j}}{\frac{\Delta x_{i+1} + \Delta x_i}{2}} = -2T_x(\Delta y_j) \frac{\phi_{i+1,j} - \phi_{i,j}}{\Delta x_{i+1} + \Delta x_i} \quad (\text{III})$$

In terms of $\phi_{i+1,j} - \phi_{i,j} = \Delta\phi$ = total head loss in series

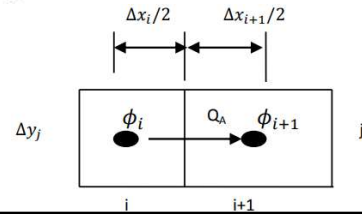
Eq. (III) is rewritten as $\Delta\phi = -\frac{Q_A}{2T_x(\Delta y_j)}(\Delta x_{i+1} + \Delta x_i)$

$$\Delta\phi = \Delta\phi_i + \Delta\phi_{i+1}$$

$$-\frac{Q_A}{2T_x(\Delta y_j)}(\Delta x_{i+1} + \Delta x_i) = -\frac{Q_A}{2T_{xi,j}(\Delta y_j)}(\Delta x_i) - \frac{Q_A}{2T_{xi+1,j}(\Delta y_j)}(\Delta x_{i+1})$$

Solving for T_x

$$T_x = \frac{T_{xi,j}T_{xi+1,j}}{\Delta x_i T_{xi+1,j} + \Delta x_{i+1} T_{xi,j}} (\Delta x_i + \Delta x_{i+1})$$



Substituting T_x in eq. (III) and finding q_A

$$q_A = \frac{Q_A}{\Delta x_i \Delta y_j} = -2T_x \frac{\phi_{i+1,j} - \phi_{i,j}}{\Delta x_i(\Delta x_{i+1} + \Delta x_i)} = -\frac{2T_{xi,j}T_{xi+1,j}}{\Delta x_i T_{xi+1,j} + \Delta x_{i+1} T_{xi,j}} (\Delta x_i + \Delta x_{i+1}) \frac{\phi_{i+1,j} - \phi_{i,j}}{\Delta x_i(\Delta x_{i+1} + \Delta x_i)}$$

$$\text{i.e., } q_A = -A_{i,j}(\phi_{i+1,j} - \phi_{i,j})$$

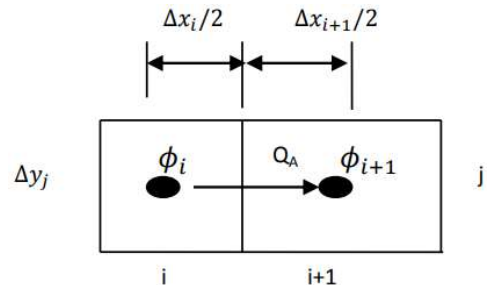
$$\text{where } A_{i,j} = \frac{2T_{xi,j}T_{xi+1,j}}{\Delta x_i(\Delta x_i T_{xi+1,j} + \Delta x_{i+1} T_{xi,j})}$$

Similarly, other Darcy fluxes in finite difference form are

$$q_C = -C_{i,j}(\phi_{i,j} - \phi_{i-1,j}) = C_{i,j}(\phi_{i-1,j} - \phi_{i,j})$$

$$q_D = -D_{i,j}(\phi_{i,j+1} - \phi_{i,j})$$

$$q_B = -B_{i,j}(\phi_{i,j} - \phi_{i,j-1}) = B_{i,j}(\phi_{i,j-1} - \phi_{i,j})$$



where

$$B_{i,j} = \frac{2T_{yi,j}T_{y,j-1}}{\Delta y_i(\Delta y_j T_{yi,j-1} + \Delta y_{j-1} T_{yi,j})}$$

$$C_{i,j} = \frac{2T_{xi,j}T_{xi-1,j}}{\Delta x_i(\Delta x_i T_{xi-1,j} + \Delta x_{i-1} T_{xi,j})}$$

$$D_{i,j} = \frac{2T_{yi,j}T_{y,j+1}}{\Delta y_j(\Delta y_i T_{yi,j+1} + \Delta y_{j+1} T_{yi,j})}$$

Substituting values of Darcy fluxes in eq. (I)

$$C_{i,j}(\phi_{i-1,j} - \phi_{i,j}) + D_{i,j}(\phi_{i,j+1} - \phi_{i,j}) + A_{i,j}(\phi_{i+1,j} - \phi_{i,j}) + B_{i,j}(\phi_{i,j-1} - \phi_{i,j}) - q_F = S \frac{\partial \phi}{\partial t} \quad (V)$$

Dropping the subscripts of coefficients

$$A\phi_{i+1,j} + B\phi_{i,j-1} + C\phi_{i-1,j} + D\phi_{i,j+1} - E\phi_{i,j} = S \frac{\partial \phi}{\partial t} + q_F \quad (VI)$$

where $E = A + B + C + D$

Simplification of homogeneous and isotropic aquifer

For homogeneous aquifer, $\Delta x = \Delta y$

For isotropic aquifer, $T_x = T_y = T$

Coefficients $(A, B, C, D) = T/\Delta x^2$, and $E = A + B + C + D = 4T/\Delta x^2$

Substituting in eq. (VI)

$$\frac{T}{\Delta x^2}(\phi_{i+1,j} + \phi_{i,j-1} + \phi_{i-1,j} + \phi_{i,j+1} - 4\phi_{i,j}) = \left(S \frac{\partial \phi}{\partial t} + q_F\right)$$

$$\text{i.e., } (\phi_{i+1,j} + \phi_{i,j-1} + \phi_{i-1,j} + \phi_{i,j+1} - 4\phi_{i,j}) = \frac{\Delta x^2}{T} \left(S \frac{\partial \phi}{\partial t} + q_F\right)$$

For steady flow with $q_F = 0$

$$\phi_{i+1,j} + \phi_{i,j-1} + \phi_{i-1,j} + \phi_{i,j+1} - 4\phi_{i,j} = 0$$

$\phi_{i,j}$ is equal to average of four neighbouring grid values.

At boundaries, four grids will not be available. So, the boundary condition should be taken into account to compute coefficients.

For impervious boundary on all sides, the coefficient from that side is zero.

i.e., for $j = 1$, $B = 0$; for $j = n$, $D = 0$; for $i = 1$, $C = 0$; and for $i = n$, $A = 0$

For grid elements adjacent to a river, only one value of space increment is needed.

e.g., $A = 2T/\Delta x^2$

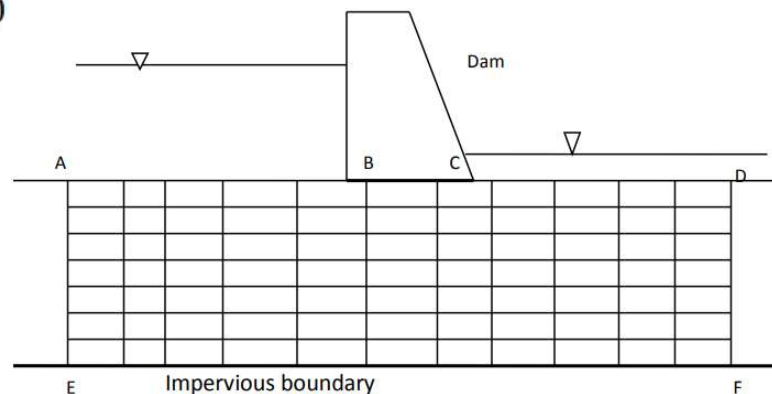
After computing ϕ at different grids, equipotential lines are drawn by interpolation and streamlines are drawn perpendicular to these lines.

7.3 Simulation of seepage under a dam based on steady state 2D model

The equation for steady-state 2D groundwater flow is

$$\frac{\partial}{\partial x} \left(T_x \frac{\partial \phi}{\partial x} \right) + \frac{\partial}{\partial y} \left(T_y \frac{\partial \phi}{\partial y} \right) = 0$$

where T_x and T_y are the transmissivity in x and y directions, respectively; and ϕ is the potential head.



Finite difference grid for seepage under dam

From eq. (VI), the continuity equation for interior grid is written as

$$A\phi_{i+1,j} + B\phi_{i,j-1} + C\phi_{i-1,j} + D\phi_{i,j+1} - E\phi_{i,j} = S \frac{\partial \phi}{\partial t} + q_F$$

For steady state condition with no withdrawal

$$A\phi_{i+1,j} + B\phi_{i,j-1} + C\phi_{i-1,j} + D\phi_{i,j+1} - E\phi_{i,j} = 0$$

$$\text{i.e., } \phi_{i,j}^+ = (A\phi_{i+1,j} + B\phi_{i,j-1} + C\phi_{i-1,j} + D\phi_{i,j+1})/E$$

where +: present iteration, without +: previous iteration

For homogeneous and isotropic medium

$$\phi_{i+1,j} + \phi_{i,j-1} + \phi_{i-1,j} + \phi_{i,j+1} - 4\phi_{i,j} = 0$$

$$\phi_{i,j}^+ = (\phi_{i+1,j} + \phi_{i,j-1} + \phi_{i-1,j} + \phi_{i,j+1})/4$$

Iterative procedure to compute potential at each grid and seepage rate

1. Consider the boundary condition for evaluating the continuity equation at boundary. In the figure, BC and EF are impervious boundaries, AB and CD are equipotential head (constant head). AE and DF can also be considered as barrier boundary if the grids extend far enough so that no flow occurs across AE and DF .
2. Assign initial value of ϕ at each grid. (usually 0 or estimated value)
3. Compute $\phi_{i,j}^+$
4. Compute $e = |\phi_{i,j}^+ - \phi_{i,j}|$

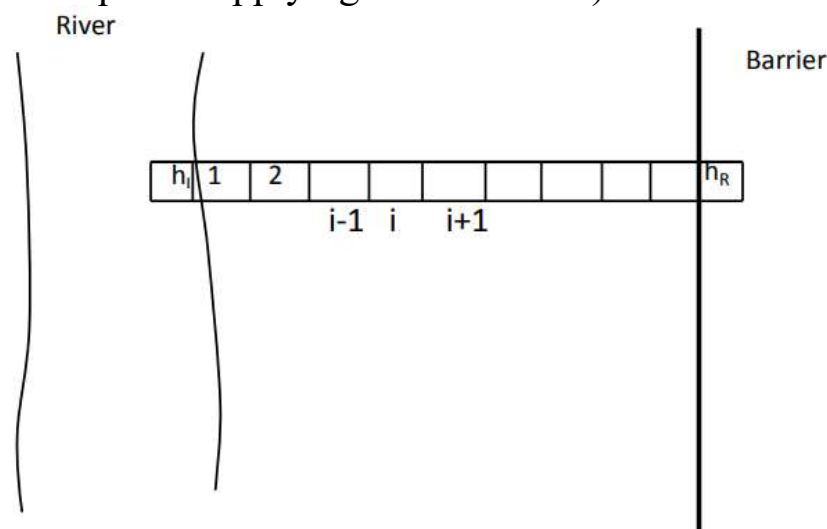
5. If e is less than some specified value, stop the iteration. Else repeat steps 3 to 4 until the condition is satisfied.

6. Compute seepage rate by using Darcy equation.

Horizontal seepage from grid $i-1$ is $Q_{i,j}^H = C_{i,j} \Delta x_i \Delta y_j (\phi_{i-1,j} - \phi_{i,j})$

Vertical seepage from grid $j-1$ is $Q_{i,j}^V = B_{i,j} \Delta x_i \Delta y_j (\phi_{i,j-1} - \phi_{i,j})$

7.4 One Dimensional implicit model for simulating river stage-water table interactions (River recharging groundwater aquifer or groundwater aquifer supplying water to river)



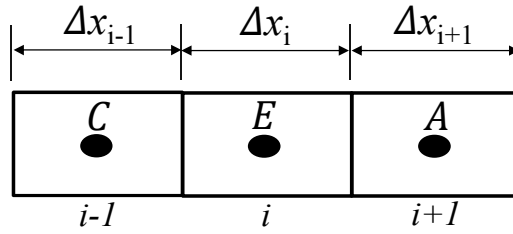


Figure: 1D finite difference grid

Partial differential equation for 1D unsteady groundwater flow is

$$\frac{\partial}{\partial x} \left(T \frac{\partial \phi}{\partial x} \right) = S \frac{\partial \phi}{\partial t} + q_F$$

where T is the transmissivity; ϕ is the water table elevation for an unconfined aquifer or the elevation of piezometric surface for confined aquifer; S is the storage coefficient; and q_F is the net groundwater withdrawal per unit area.

The discrete form of continuity equation for grid i in terms of h is written as (similar to 2D model with exclusions of terms in y direction)

$$A_i(h_{i+1}^+ - h_i^+) + C_i(h_{i-1}^+ - h_i^+) = \frac{S_i}{\Delta t}(h_i^+ - h_i) + q_F$$

where h and h^+ are the water table elevation at the beginning and end of time step Δt . The difference equation in LHS is in terms of unknown values at time step $t+1$. Therefore, it is implicit scheme.

Here
$$A_i = \frac{2T_{xi}T_{xi+1}}{\Delta x_i(\Delta x_i T_{xi+1} + \Delta x_{i+1} T_{xi})}, C_i = \frac{2T_{xi}T_{xi-1}}{\Delta x_i(\Delta x_i T_{xi-1} + \Delta x_{i-1} T_{xi})}$$

Now,
$$A_i h_{i+1}^+ + C_i h_{i-1}^+ - \left(A_i + C_i + \frac{S_i}{\Delta t} \right) h_i^+ = -\frac{S_i}{\Delta t} h_i + q_F$$

Defining
$$E_i = -\left(A_i + C_i + \frac{S_i}{\Delta t} \right) \text{ and } Q_i = -\frac{S_i}{\Delta t} h_i + q_F$$

For homogeneous (same value of for all grids) and isotopic (same value of T for all grids),

$$A_i = C_i = \frac{T}{\Delta x^2} \text{ and } E_i = -\left(\frac{2T}{\Delta x^2} + \frac{S_i}{\Delta t}\right)$$

The continuity equation becomes

$$C_i h_{i-1}^+ + E_i h_i^+ + A_i h_{i+1}^+ = Q_i$$

Writing this equation for each grid results in tri-diagonal coefficient matrix

$$\begin{bmatrix} E_1 & A_1 & 0 & 0 & 0 & 0 & 0 \\ C_2 & E_2 & A_2 & 0 & 0 & 0 & 0 \\ 0 & C_3 & E_3 & A_3 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 0 & C_n & E_n \end{bmatrix} \begin{bmatrix} h_1^+ \\ h_2^+ \\ h_3^+ \\ \vdots \\ h_n^+ \end{bmatrix} = \begin{bmatrix} Q_1 - C_1 h_L \\ Q_2 \\ Q_3 \\ \vdots \\ Q_n - A_n h_R \end{bmatrix}$$

where h_L is the river stage at potential boundary

As the right boundary is barrier boundary, $A_n = 0$.

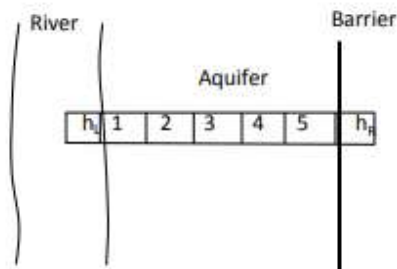
The system of linear equations is solved to find h .

The seepage rate between grid elements is computed by

$$Q_i = C_i \Delta x_i (h_{i-1}^+ - h_i^+)$$

Loss or gain at the river = seepage at grid element I (i.e., $i = I$).

A schematic for simulating river stage-water table fluctuation is shown in figure below.



The following data are given for the simulation: homogeneous and isotropic aquifer, river stage (h_i) = 203m, Length of aquifer = 500m, $\Delta x = 100$ m, $\Delta t = 1$ day, transmissivity of aquifer = 0.03m/s, storage coefficient = 0.01. The initial value of water table at 5 grids given as 99.1, 99.11, 99.2, 99.3, 99.4m respectively. Compute coefficients of the 1D implicit finite difference model and display the matrix of known and unknowns.

Solution:

The finite difference equation

$$C_i h_{i-1}^+ + E_i h_i^+ + A_i h_{i+1}^+ = Q_i$$

h and h^+ = water table elevation at the beginning and end of time step Δt

For homogeneous and isotropic aquifer, coefficients are computed as

$$A_i = C_i = \frac{T}{\Delta x^2} = \frac{0.03}{100^2} = 3 \times 10^{-6} \text{ (Same for all grids)}$$

$$E_i = -\left(\frac{2T}{\Delta x^2} + \frac{S_i}{\Delta t}\right) = -\left(\frac{2 \times 0.03}{100^2} + \frac{0.01}{1 \times 24 \times 3600}\right) = -7 \times 10^{-6} \text{ (Same for all grids)}$$

Computation of RHS part

$$Q_i = -\frac{S_i}{\Delta t} h_i + q_F$$

Take $q_F = 0$

$$Q_i = -\frac{S_i}{\Delta t} h_i \text{ where } h_i = \text{initial values}$$

$$Q_1 = -\frac{0.01}{1 \times 24 \times 3600} 99.1 = 1.14 \times 10^{-5}$$

$$Q_2 = -\frac{0.01}{1 \times 24 \times 3600} 99.11 = 1.147 \times 10^{-5}$$

$$Q_3 = -\frac{0.01}{1 \times 24 \times 3600} 99.2 = 1.148 \times 10^{-5}$$

$$Q_4 = -\frac{0.01}{1 \times 24 \times 3600} 99.3 = 1.149 \times 10^{-5}$$

$$Q_5 = -\frac{0.01}{1 \times 24 \times 3600} 99.4 = 1.15 \times 10^{-5}$$

Boundary conditions

$$h_L = h_0 = 203\text{m}$$

$$\text{For } i = 1: C_1 h_0^+ + E_1 h_1^+ + A_1 h_2^+ = Q_1$$

$$E_1 h_1^+ + A_1 h_2^+ = Q_1 - C_1 h_L$$

$$\text{For } i = 2: C_2 h_1^+ + E_2 h_2^+ + A_2 h_3^+ = Q_2$$

$$\text{For } i = 3: C_3 h_2^+ + E_3 h_3^+ + A_3 h_4^+ = Q_3$$

$$\text{For } i = 4: C_4 h_3^+ + E_4 h_4^+ + A_5 h_5^+ = Q_4$$

$$\text{For } i = 5: C_5 h_4^+ + E_5 h_5^+ + A_5 h_6^+ = Q_5$$

$$C_5 h_4^+ + E_5 h_5^+ + 0 = Q_5$$

Coefficient matrix is 5x5 matrix (tridiagonal elements non-zero, others zero)

$$\begin{bmatrix} E_1 & A_1 & 0 & 0 & 0 \\ C_2 & E_2 & A_2 & 0 & 0 \\ 0 & C_3 & E_3 & A_3 & 0 \\ 0 & 0 & C_4 & E_4 & A_4 \\ 0 & 0 & 0 & C_5 & E_5 \end{bmatrix} \begin{bmatrix} h_1^+ \\ h_2^+ \\ h_3^+ \\ h_4^+ \\ h_5^+ \end{bmatrix} = \begin{bmatrix} Q_1 - C_1 h_L \\ Q_2 \\ Q_3 \\ Q_4 \\ Q_5 \end{bmatrix}$$

$$\begin{bmatrix} -7 \times 10^{-6} & 3 \times 10^{-6} & 0 & 0 & 0 \\ 3 \times 10^{-6} & -7 \times 10^{-6} & 3 \times 10^{-6} & 0 & 0 \\ 0 & 3 \times 10^{-6} & -7 \times 10^{-6} & 3 \times 10^{-6} & 0 \\ 0 & 0 & 3 \times 10^{-6} & -7 \times 10^{-6} & 3 \times 10^{-6} \\ 0 & 0 & 0 & 3 \times 10^{-6} & -7 \times 10^{-6} \end{bmatrix} \begin{bmatrix} h_1^+ \\ h_2^+ \\ h_3^+ \\ h_4^+ \\ h_5^+ \end{bmatrix} = \begin{bmatrix} 5.98 \times 10^{-4} \\ 1.147 \times 10^{-5} \\ 1.148 \times 10^{-5} \\ 1.149 \times 10^{-5} \\ 1.150 \times 10^{-5} \end{bmatrix}$$